

VISITING EARLY AT PRIME TIMES

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ABSTRACT. Given an integer $m \geq 2$ and a sufficiently large q , we prove that there always exists an arithmetic progression with common difference q for which the m -th least prime in such progression is $\ll q \exp(3.8075m)$. This is obtained using a minorant for the indicator of primes in the Maynard–Tao sieve, in parallel with Baker–Irving and Stadlmann’s work on bounded gaps between primes. The dependence on q is best possible. Furthermore, we generalize our result to dynamical systems. The quality of the result depends crucially on the first return time, which we illustrate in the case of Diophantine approximation.

1. INTRODUCTION

Analytic Number Theory has witnessed many breakthroughs on fundamental problems in recent years, in particular the bounded gaps between primes, thanks to Zhang [Zha14], Maynard [May15] and Tao (see also [Pol14b]), building upon the celebrated work of Goldston–Pintz–Yıldırım [GPY09]. Maynard [May16] further proved that any subset of primes which is “well distributed” in arithmetic progressions contains many primes that are close together.

In particular, for primes in a fixed progression, the best known result is due to Baker and Zhao [BZ16], who proved the following. Given $m \geq 1$, let x be a sufficiently large real number, $q = x^\theta$ a suitably “powerful” modulus with $0 < \theta \leq 5/12 - \eta$ for a fixed $\eta > 0$ and $\prod_{p|q} p \leq (\log x)^C$ for a fixed $C > 0$, and $(a, q) = 1$. Then there exist primes $p_1 < \dots < p_m$ in $(x/2, x]$ such that $p_i \equiv a \pmod{q}$ for $1 \leq i \leq m$ and

$$p_m - p_1 \leq q \exp(cm)$$

for some constant $c = c(\theta) > 0$.

Apart from progressions, there are also infinitely many bounded gaps between primes in a fixed “Chebotarev set” [Tho14], a suitable “Sato–Tate interval” [Gil+20] or a “nil–Bohr set” [ST21].

To motivate our work, we begin by asking: how big is the first prime in a given arithmetic progression? Let $p(q, a)$ denote the least prime in the arithmetic progression $a \pmod{q}$. Then recall Linnik’s celebrated theorem that $p(q, a) \ll q^L$ for some absolute constant $L > 0$. If $p_m(q, a)$ is now the m^{th} prime in the arithmetic progression $a \pmod{q}$, then a quantitative version of Linnik’s theorem [IK04, Corollary 18.8] implies

$$\max_{\substack{a \pmod{q} \\ (a, q) = 1}} p_m(q, a) \ll_m q^L,$$

for some $L > 0$.

Assuming a uniform variant of the Hardy–Littlewood prime k -tuple conjecture, the second author [Leu26] showed that as soon as $f(q) \rightarrow \infty$ with $q \rightarrow \infty$, the least prime $p(q, a)$ is $\leq f(q)\varphi(q) \log q$ for almost all $a \pmod{q}$. In fact, the least prime in arithmetic progressions to a common large modulus has an exponential limiting distribution. More

precisely, if a reduced residue $a \pmod{q}$ is chosen uniformly at random, then we have the convergence in distribution

$$\frac{p(q, a)}{\varphi(q) \log q} \xrightarrow{d} \text{Exp}(1) \quad \text{as } q \rightarrow \infty.$$

Given $m \geq 1$ and $(a, q) = 1$, we are also interested in the m -th least prime $p \equiv a \pmod{q}$, which we denote as $p_m(q, a)$. For any residue class containing fewer than m primes, we set $p_m(q, a) = \infty$. As $a \pmod{q}$ varies uniformly with $q \rightarrow \infty$, the second author [Leu25] further showed conditionally that the normalized primes in the progression,

$$\frac{p_1(q, a)}{\varphi(q) \log q} < \frac{p_2(q, a)}{\varphi(q) \log q} < \dots,$$

are located on the positive real number line $(0, \infty)$ as if they were randomly positioned points, in the sense of a Poisson point process.

In this paper, we study $\min_{a \pmod{q}} p_m(q, a)$. By the pigeonhole principle, one can immediately show that $\min_{a \pmod{q}} p_m(q, a) \ll_m \varphi(q) \log q$. Applying a variant of the Maynard–Tao sieve weights, we improve this bound.

Theorem 1.1. *For $m \geq 2$ and q sufficiently large in terms of m , we have*

$$\min_{a \pmod{q}} p_m(q, a) \ll q \cdot \exp(3.8075m).$$

Remark 1.1. A similar result can be obtained from the setting of Maynard [May16, Theorem 3.1], but this natural statement does not seem to have been observed before in the literature. Moreover, Maynard’s theorem gives the upper bound $O(qe^{cm})$ for some large constant c . If we go into the proof and optimise the constant for our special case, we may take any $c > 4$. As a by-product of our result, we see that we can take the exponent $c = 3.8075$ like in Stadlmann [Sta25].

When keeping m fixed and letting q grow, the order of magnitude is optimal. Indeed, in any arithmetic progression, the m -th prime is at least the m -th positive integer in that progression. Thus $\min_{a \pmod{q}} p_m(q, a) > (m-1)q$.

The dependence on m in Theorem 1.1 could be improved under the following variant of the Hardy–Littlewood conjecture: For all fixed $\varepsilon > 0$ and fixed admissible m -tuple of minimal diameter $\mathcal{H} = \{0 = h_1 < \dots < h_m\}$, we expect that for sufficiently large q , we must have

$$\#\left\{\varepsilon q < a < 2\varepsilon q : a + qh_i \text{ is prime for all } i\right\} \sim \mathfrak{S}(\mathcal{H}, q) \frac{\varepsilon q}{(\log q)^m}$$

with

$$\mathfrak{S}(\mathcal{H}, q) = \left(\frac{q}{\varphi(q)}\right)^{m-1} \prod_{p \nmid q} \left(1 - \frac{\#\{h_i \pmod{p}\}}{p}\right) \left(1 - \frac{1}{p}\right)^{-m}.$$

This variant of the Hardy–Littlewood conjecture is consistent with the Cramér–Granville model (see [Gra95]). With this conjecture, we directly obtain

$$\min_{a \pmod{q}} p_m(q, a) < (H(m) + o(1))q$$

as we let $q \rightarrow \infty$, with $H(k)$ being the diameter of the narrowest admissible k -tuple. Since $H(m) \asymp m \log m$, we would have

$$\min_{a \pmod{q}} p_m(q, a) \ll qm \log m, \tag{1.1}$$

conditional on the variant of the Hardy–Littlewood conjecture.

The bound (1.1) would be sharp: Consider a, q satisfying $(a, q) = 1$ and a prime modulus $q > m$. If $(a + \ell_i q)_{i=1}^m$ are the first m primes in the arithmetic progression $a \pmod{q}$, then the set $\{\ell_2, \dots, \ell_m\}$ is admissible. Therefore, taking the minimum over all integers a coprime to q , we have

$$\min_{a \pmod{q}} p_m(q, a) > (H(m-1) + 1)q.$$

Also, by varying the progressions instead of fixing a particular one, we are able to obtain a prime cluster early,¹ which makes the nature of our result distinct from the existing literature, including those discussed in the beginning of the paper.

Furthermore, we generalize our result to dynamical systems, for example circle rotations (Diophantine approximation, see Example 2.1). It turns out that the quality of the result depends crucially on the first return time, which we explore in Section 7. In particular, in the case of circle rotations $T_\alpha : x \mapsto x + \alpha$ for almost all α or those α with bounded partial quotients, we establish tighter lower and upper bounds than those in the literature for the first return time (see Proposition 7.1).

Recurrence is a central theme in ergodic theory, i.e., how points in dynamical systems return close to themselves under iteration. In 1890, Poincaré proved a simple yet far-reaching recurrence theorem in the context of the “three-body” problem of planetary orbits (see [EW11, p. 21]). For our purposes, here we state a simple version (see [EW11, Exercise 2.2.1] for instance).

Lemma 1.1 (Poincaré recurrence theorem). *Given a measure-preserving system $(X, \mathcal{B}, \mu, T)$ with $\mu(X) = 1$, let $A \in \mathcal{B}$ with $\mu(A) > 0$. Then there exists an integer $1 \leq n \leq \lfloor \mu(A)^{-1} \rfloor$ such that*

$$\mu(A \cap T^{-n}A) > 0.$$

Indeed, if $\mu(A \cap T^{-n}A) = 0$ for $1 \leq n \leq N := \lfloor \mu(A)^{-1} \rfloor$, then $A, T^{-1}A, \dots, T^{-N}A$ are pairwise disjoint up to null sets, contradicting $(N+1)\mu(A) > 1$. A famous example is Dirichlet’s approximation theorem, corresponding to circle rotation. Inspired by the recurrence theorem, we shall visit early at prime times in *metric-measure-preserving systems* defined as follows.

Definition 1.1 (Metric-measure-preserving system). A *metric-measure-preserving system* is a quintuple $(X, d, \mathcal{B}, \mu, T)$ where (X, d) is a metric space and $(X, \mathcal{B}, \mu, T)$ is a measure-preserving system satisfying the following compatibility conditions:

- the Borel σ -algebra $\mathcal{B}$ is induced by the metric d ;
- the probability measure μ is *pseudo-doubling*, i.e., there exist constants $\lambda \geq 1$ and $r_0 > 0$ such that

$$\mu(B(x; 2r)) \leq \lambda \cdot \mu(B(x; r))$$

for any $x \in X$ and $0 < r < r_0$, where $B(x; r) := \{y \in X : d(y, x) < r\}$;

- the measure-preserving map T is an isometry.

Let $p_m^T(x_0, x; \varepsilon)$ denote the m -th least prime p for which $T^p x_0 \in B(x; \varepsilon)$. If fewer than m primes satisfy this condition, we set $p_m^T(x_0, x; \varepsilon) = \infty$. Then, with the definition in hand, we state the dynamical generalization of Theorem 1.1.

¹See [Pom80] and a recent improvement [LPS17] for a lower bound for the least prime in a progression instead.

Theorem 1.2. *Let $m \geq 2$ and $(X, d, \mathcal{B}, \mu, T)$ be a metric-measure-preserving system. Then for $0 < \varepsilon < r_0$ sufficiently small (in terms of m) and $x_0 \in \text{supp } \mu$, we have*

$$\min_{x \in X} p_m^T(x_0, x; \varepsilon) \ll_{m, \lambda} \mu(B(x_0; \varepsilon))^{-1}.$$

Notation. Throughout the paper, we use the standard big O and little o notations as well as the Vinogradov notations $\ll, \gg$ and the Hardy notation $\asymp$, where the implied constants depend only on the subscripted parameters.

2. MAIN RESULTS

We start by defining families of minorants for the indicator function of prime numbers. This definition is inspired by a similar one given by Baker and Irving in [BI17]. For $n \geq 2$, let $P^-(n)$ denote the least prime factor of n , and set $P^-(1) = \infty$.

Definition 2.1 (Minorants for the indicator of primes). Fix a positive integer ℓ , parameters $(\theta, \xi, \beta) \in (0, 1]^3$, and a function $\delta: (0, \theta) \rightarrow (0, \xi]$. We define $\text{BI}_\ell[\theta, \xi, \beta, \delta]$ as the set of functions $\rho: \mathbb{N} \times (1, \infty) \rightarrow \mathbb{R}$ satisfying

- (i) There exists $C > 0$ such that $\rho(a, q) \leq 1_{\mathbb{P}}(a)$ for every $a > q > 1$, and $-C \leq \rho(a, q)$ whenever $q > 1$ and $\ell q < a \leq (\ell + 1)q$.
- (ii) For every a, q satisfying $a > q > 1$ and $\rho(a, q) \neq 0$, we have $P^-(a) \geq q^\xi$.
- (iii) For every $q \geq 2$, $c \in \mathbb{Z}$, $A > 0$ and $\varepsilon \in (0, \theta)$, we have

$$\sum_{\substack{r \leq q^{\theta-\varepsilon} \\ r|P}} \left| \sum_{\ell q < a \leq (\ell+1)q} \rho(a, q) \left(1_{a \equiv c \pmod{r}} - \frac{1_{(a,r)=1}}{\varphi(r)} \right) \right| \ll_{A, \varepsilon} \frac{q}{(\log q)^A}$$

with $P = \prod_{\substack{p \leq q^{\delta(\varepsilon)} \\ p \nmid c}} p$.

- (iv) We have

$$\sum_{\ell q < a \leq (\ell+1)q} \rho(a, q) \geq (\beta - o(1)) \frac{q}{\log q}$$

as $q \rightarrow \infty$.

Baker and Irving [BI17, Lemma 1] proved the following theorem on bounded gaps between primes: If $\text{BI}_1[\theta, \xi, \beta, \delta]$ is not empty for some parameters $(\theta, \xi, \beta) \in (0, 1]^3$ and some function $\delta: (0, \theta) \rightarrow (0, \xi]$, then

$$H_m := \liminf_{n \rightarrow \infty} (p_{n+m} - p_n) \ll_\varepsilon \exp\left(\frac{(2 + \varepsilon)m}{\theta\beta}\right) \quad (2.1)$$

where $p_1 < p_2 < \dots$ is the increasing sequence of prime numbers.

Different theorems in the literature are proofs that the indicator $1_{\mathbb{P}}$ is in $\text{BI}_1[\theta, 1, 1, \delta]$ for some θ and δ , leading to proofs of the form $H_m \ll_\varepsilon e^{(2/\theta + \varepsilon)m}$:

- The Bombieri-Vinogradov Theorem implies that we can take $\theta = \frac{1}{2}$ and the function $\delta \equiv 1$. We note that Maynard [May15] proved that $H_m \ll m^3 e^{4m}$. The exponent 4 comes from applying the Bombieri-Vinogradov Theorem in a key part of the proof.
- Zhang [Zha14] essentially proved that we can take $\theta = \frac{1}{2} + \frac{1}{584}$ and any function δ satisfying $\delta(\varepsilon) \in (0, \frac{1}{1168})$.
- Polymath [Pol14a] proved that we can take $\theta = \frac{1}{2} + \frac{7}{300}$ and any δ satisfying $\delta(\varepsilon) < \frac{5\varepsilon}{3}$.
- Stadlmann [Sta25] proved that we can take $\theta = \frac{1}{2} + \frac{1}{40}$ and any δ satisfying $\delta(\varepsilon) < \frac{3\varepsilon}{2}$.

With the Elliott-Halberstam conjecture, we expect that $1_{\mathbb{P}} \in \bigcap_{\ell \geq 1} \text{BI}_{\ell}[1, 1, 1, 1]$. (It is important to note here that when taking the intersection, the implicit constant in Definition 2.1 (iii), as well as the speed of convergence for Definition 2.1 (iv) are not necessarily the same for all $\ell \geq 1$.)

Concerning non-trivial minorants of $1_{\mathbb{P}}$, we have that

- Baker and Irving [BI17] showed that for all $\varepsilon > 0$, there exist ξ and δ such that $\text{BI}_1\left[\frac{1}{2} + \frac{16}{659} - \varepsilon, \xi, 1 - 8 \cdot 10^{-6}, \delta\right]$ is not empty.
- Stadlmann [Sta25] constructed the following minorant:

$$\rho_0(a, q) := 1_{\mathbb{P}}(a) - \sum_{\substack{a=p_1 p_2 p_3 p_4 n_5 \\ 0.19038 \leq \alpha_4 < \alpha_3 < \alpha_2 < \alpha_1 \leq 0.40481 \\ \alpha_1 + \alpha_2 < 0.40481 \\ \alpha_2 + \alpha_3 + \alpha_4 > 0.59519 \\ P^-(n_5) \geq p_4}} 1 - \sum_{\substack{a=p_2 p_3 p_4 p_5 p_6 \\ 0.19038 \leq \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6 \leq 0.23848 \\ \alpha_2, \alpha_4 > \alpha_3 \\ \alpha_2 + \alpha_4 < 0.40481 \\ \alpha_2 + \alpha_3 + \alpha_5 > 0.59519 \\ \alpha_6 \geq \alpha_5}} 1, \quad (2.2)$$

where $\alpha_j = \frac{\log p_j}{\log q}$ in the sums above. She proved that for $\delta \equiv 10^{-10}$, we have $\rho_0 \in \text{BI}_1[0.5253, 0.19038, 0.99998, \delta]$.

We can now state an analogous statement to Baker and Irving's result (2.1) for primes in arithmetic progressions, which provides a lower bound for the number of residue classes with early prime clusters.

Theorem 2.1. *Fix $\varepsilon > 0$ and let $m \geq 2$. Suppose that $\bigcap_{\ell \geq 1} \text{BI}_{\ell}[\theta, \xi, \beta, \delta]$ is not empty for some parameters $(\theta, \xi, \beta) \in (0, 1]^3$ and some function $\delta: (0, \theta) \rightarrow (0, \xi]$ satisfying $\delta(y) \gg y$. Then for any sufficiently large modulus q (in terms of m and ε), there exist constants $C, C' > 0$ which depend on $(\theta, \xi, \beta, \delta, \varepsilon)$ such that*

$$\#\left\{a \pmod{q} : p_m(q, a) \leq Cq \exp\left(\frac{2}{\theta\beta}(1 + \varepsilon)m\right)\right\} \gg_{m, \varepsilon} \varphi(q)(\log q)^{-C' \exp\left(\frac{2}{\theta\beta}(1 + \varepsilon)m\right)}.$$

Now, we can state our theorem in the case of Stadlmann's minorant. For this, we have the following lemma.

Lemma 2.1. *Let ρ_0 be defined as in (2.2). For the constant function $\delta(\varepsilon) = 10^{-10}$, we have*

$$\rho_0 \in \bigcap_{\ell \geq 1} \text{BI}_{\ell}[0.5253, 0.19038, 0.99998, \delta].$$

Proof. Properties (i) and (ii) of Definition 2.1 follow directly from the definition of ρ_0 . Concerning properties (iii) and (iv), Stadlmann proved in [Sta25, Section 4] that $\rho_0 \in \text{BI}_{\ell}[0.5253, 0.19038, 0.99998, \delta]$ for $\ell = 1$. For each fixed $\ell \geq 2$, we apply the same decomposition and distribution estimates on $(\ell q, (\ell + 1)q]$. Since $\log(\ell q) = \log q + O_{\ell}(1)$, replacing $\log q$ by $\log(\ell q)$ changes each exponent $\alpha_j = \log p_j / \log q$ by $O_{\ell}(1/\log q)$. For fixed ℓ , these $O_{\ell}(1/\log q)$ perturbations are eventually smaller than the fixed slack in the Type I/II/III ranges used in Stadlmann's argument, so the same distribution estimates remain applicable after slightly enlarging the corresponding ranges. Their constants, thresholds, and $o(1)$ rates may depend on ℓ , while the numerical parameters 0.5253, 0.19038, 0.99998 and $\delta \equiv 10^{-10}$ remain unchanged. $\square$

Using this, we get

Corollary 2.1. *We have for q sufficiently large in terms of m ,*

$$\#\{a \pmod{q} : p_m(q, a) \leq Cq \exp(3.8075m)\} \gg_m \varphi(q)(\log q)^{-C' \exp(3.8075m)}.$$

We also state the quantitative version of Theorem 1.2, which is the dynamical generalization of Theorem 2.1.

Theorem 2.2. *Let $m \geq 2$ and $(X, d, \mathcal{B}, \mu, T)$ be a metric-measure-preserving system. Then there exist constants $C = C(\lambda), C' = C'(\lambda) > 0$ such that for any nonperiodic $x_0 \in \text{supp } \mu$ and $0 < \varepsilon < r_0$ sufficiently small (in terms of m and x_0), we have*

$$\begin{aligned} & \mu(\{x \in X : p_m^T(x_0, x; \varepsilon) \leq Cq \exp(3.8075m)\}) \\ & \gg_{m, \lambda} \varphi(q)(1 + \log q)^{-C' \exp(3.8075m)} \mu(B(x_0; \varepsilon)) \end{aligned}$$

for some integer $1 \leq q \ll_{\lambda} m^{\log_2 \lambda} \exp(3.8075 \cdot m \log_2 \lambda) \mu(B(x_0; \varepsilon))^{-1}$.

Remark 2.1. In the proof of Theorem 2.2, q is the first return time of x_0 to $B(x_0; \varepsilon/(2h))$, with h defined there.

Remark 2.2. We could also state our theorem in the style of Theorem 2.1 using the minorants, but we omit this for the sake of clarity.

We provide an example to illustrate this theorem.

Example 2.1 (Circle rotation). *Consider the metric-measure-preserving system with $X = \mathbb{R}/\mathbb{Z}, d = \|\cdot\|, \mu$ being the normalized Haar measure and $T_{\alpha} : x \mapsto x + \alpha \pmod{1}$, where $\|\cdot\|$ is the distance to the nearest integer function and α irrational in $[0, 1]$. Let $m \geq 2$. Then for any sufficiently small $\varepsilon > 0$ (in terms of m), we have*

$$\min_{\beta \in [0, 1]} p_m^{\alpha}(\beta; \varepsilon) \ll \varepsilon^{-1} m \exp(7.615m)$$

where $p_m^{\alpha}(\beta; \varepsilon)$ denotes the m -th least prime p for which $\|p\alpha - \beta\| < \varepsilon$. If fewer than m primes satisfy this condition, we set $p_m^{\alpha}(\beta; \varepsilon) = \infty$. Moreover, for a typical irrational $\alpha \in [0, 1]$ and $\varepsilon > 0$ sufficiently small in terms of m and α , we can establish a more precise version of Theorem 2.2 in this case by showing that the first return time is $\asymp \varepsilon^{-1+o(1)}$; see Section 7 and Proposition 7.1 for details.

3. A VARIANT OF MAYNARD–TAO SIEVE WEIGHT

In Sections 3–5, μ denotes the Möbius function; in the dynamical sections, it denotes the probability measure.

Let $m, k \geq 2$. Let $(h_1, \dots, h_k)$ be an admissible k -tuple, i.e., a k -tuple of non-negative integers satisfying $0 \leq h_1 < \dots < h_k$ and $\#\{h_i \pmod{p} : 1 \leq i \leq k\} < p$ for any prime p . Fix also θ, ξ, β and δ such that $\bigcap_{\ell \geq 1} \text{BI}_{\ell}[\theta, \xi, \beta, \delta]$ is not empty.

Then, we would like to construct a sequence of non-negative weights w_a over the reduced residue system $\{a \in (q, 2q] : (a, q) = 1\}$ for any $q = q(m)$ sufficiently large, so that

$$S := \sum_{\substack{q < a \leq 2q \\ (a, q) = 1}} \left(\sum_{i=1}^k \rho(a + qh_i, q) - (m-1) - k \sum_{i=1}^k \sum_{\substack{p|a+qh_i \\ p \leq q^{\gamma}}} 1 \right) w_a \quad (3.1)$$

is positive for some small $\gamma = \gamma(k) > 0$ and a minorant $\rho \in \bigcap_{\ell \geq 1} \text{BI}_{\ell}[\theta, \xi, \beta, \delta]$. Let $\Omega(n)$ count the prime factors of n with multiplicity. We define

$$y_a := \sum_{i=1}^k 1_{\mathbb{P}}(a + qh_i) - (m-1) - k \sum_{i=1}^k \sum_{\substack{p|a+qh_i \\ p \leq q^{\gamma}}} 1 \quad (3.2)$$

$$M(z) := \max \left\{ w_a : a \in (q, 2q] \text{ satisfying } \Omega(a + qh_i) < z \text{ for all } i \right\}$$

Suppose that $q \geq 2 + h_k$. If $y_a \geq 1$, then the finite sequence $(a + qh_i)_{i=1}^k$ must contain at least m primes and all terms of the sequence must satisfy $P^-(a + qh_i) > q^{\gamma}$. It therefore

implies that $\Omega(a + qh_i) < 2/\gamma$ for all i . Thus, we obtain the following lower bound

$$\begin{aligned} & \#\left\{a \in (q, 2q] : \text{at least } m \text{ of the } a + qh_i \text{ are primes}\right\} \\ & \geq \sum_{\substack{q < a \leq 2q \\ (a,q)=1 \\ y_a \geq 1}} 1 \geq \sum_{\substack{q < a \leq 2q \\ (a,q)=1 \\ y_a \geq 1}} \left(\frac{y_a}{k}\right) \cdot \left(\frac{w_a}{M(2/\gamma)}\right) \geq \frac{S}{k \cdot M(2/\gamma)}. \end{aligned} \quad (3.3)$$

To construct our weight w_a , let $w := \log \log \log q$ and $W_q := \prod_{\substack{p \leq w \\ p \nmid q}} p$. By the admissibility of the k -tuple $(h_1, \dots, h_k)$, we can choose $b_0 \pmod{W_q}$ so that for those $a \equiv b_0 \pmod{W_q}$, we have $a + qh_i \equiv b_0 + qh_i \not\equiv 0 \pmod{p}$ for any prime $p \leq w, p \nmid q$. Then, analogously to the Maynard–Tao sieve weight (see [Pol14b, p.11]), we let

$$w_a := 1_{a \equiv b_0 \pmod{W_q}} \left(\sum_{\substack{d_i | a + qh_i \\ i=1, \dots, k}} \cdots \sum_{i=1}^k \prod_{i=1}^k \mu(d_i) \times F\left(\frac{\log d_1}{\log q}, \dots, \frac{\log d_k}{\log q}\right) \right)^2 \quad (3.4)$$

for some function $F : [0, \infty)^k \rightarrow \mathbb{R}$ supported on the truncated simplex

$$\Delta_k(\theta, \delta) := \left\{ (t_1, \dots, t_k) \in [0, \delta(\frac{\theta}{4 \log k})]^k : t_1 + \dots + t_k \leq \frac{\theta}{2} \left(1 - \frac{1}{2 \log k}\right) \right\}.$$

We require F to be a finite linear combination of smooth tensor products $\prod_{i=1}^k f_i(t_i)$, with $f_i \in \mathcal{C}_c^\infty([0, \infty))$, such that each tensor product has support contained in the interior of $\Delta_k(\theta, \delta)$ taken within $[0, \infty)^k$. For future reference, the family of such functions is denoted as $\mathcal{F}_k(\theta, \delta)$.

With our choice of weight w_a , it turns out that not only is the sum S in (3.1) computable, but it is also positively large.

To simplify notation, given any square-integrable function $G : [0, \infty)^k \rightarrow \mathbb{R}$, let us denote the singular integrals

$$\mathfrak{I}(G) := \int_{[0, \infty)^k} G(t_1, \dots, t_k)^2 dt_1 \cdots dt_k$$

and

$$\mathfrak{I}_i(G) := \int_{[0, \infty)^{k-1}} \left(\int_{[0, \infty)} G(t_1, \dots, t_k) dt_i \right)^2 dt_1 \cdots dt_{i-1} dt_{i+1} \cdots dt_k$$

for $1 \leq i \leq k$. Then in view of (3.1), we are required to estimate the following sums.

Proposition 3.1 (Non-prime sum). *Fix $k \geq 2$, $\theta \in (0, 1]$, $\delta : (0, \theta) \rightarrow (0, 1]$, and $F \in \mathcal{F}_k(\theta, \delta)$. We define w_a as in (3.4). Then as we let $q \rightarrow \infty$, we have*

$$\sum_{\substack{q < a \leq 2q \\ (a,q)=1}} w_a = (\mathfrak{I}(\partial_{t_1 \dots t_k} F) + o(1)) \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}.$$

Proposition 3.2 (Prime sum). *Fix $k \geq 2$, $1 \leq i_0 \leq k$, $(\theta, \xi, \beta) \in (0, 1]^3$, $\delta : (0, \theta) \rightarrow (0, \xi]$ and $F \in \mathcal{F}_k(\theta, \delta)$. We define w_a as in (3.4). Suppose that the set $\bigcap_{\ell \geq 1} \text{BI}_\ell[\theta, \xi, \beta, \delta]$ is not empty, and fix ρ , an element of this set. Then as $q \rightarrow \infty$, we have*

$$\sum_{\substack{q < a \leq 2q \\ (a,q)=1}} \rho(a + qh_{i_0}, q) w_a \geq (\beta \cdot \mathfrak{I}_{i_0}(\partial_{t_1 \dots t_k} F) - o(1)) \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}.$$

Proposition 3.3 (Small prime factor sum). *Fix $k \geq 2$, $1 \leq i_0 \leq k$, and $F \in \mathcal{F}_k(\theta, \delta)$ for some $\theta \in (0, 1]$ and $\delta: (0, \theta) \rightarrow (0, 1]$. Let $0 < \gamma \leq \frac{1}{100k}$ and $q \geq 2$. Then*

$$\sum_{\substack{q < a \leq 2q \\ (a, q) = 1}} \left(\sum_{\substack{p|a+qh_{i_0} \\ p \leq q^\gamma}} 1 \right) w_a \ll_F \gamma \times \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}.$$

4. PROOF OF THEOREM 2.1

In this section, we shall prove Theorem 2.1 assuming the aforementioned propositions. All the implicit constants in this section are allowed to depend on F, θ, ξ, β and δ .

We start by fixing $(\theta, \xi, \beta) \in (0, 1]^3$ and $\delta: (0, \theta) \rightarrow (0, \xi]$ such that the intersection $\bigcap_{\ell \geq 1} \text{BI}_\ell[\theta, \xi, \beta, \delta]$ is non-empty, and we choose ρ as an element of this intersection. Put

$$R_k := \frac{\theta}{2} \left(1 - \frac{1}{2 \log k} \right).$$

Since $\delta(y) \gg y$, choose $d > 0$ such that $\delta(y) \geq dy$ for all sufficiently small y , and fix $0 < b < d/2$. The argument proving [Pol14b, Theorem 40] gives, for all sufficiently large k , a nonzero function H_k supported in

$$\{(u_1, \dots, u_k) \in [0, b/\log k]^k : u_1 + \dots + u_k \leq 1\}$$

such that $\sum_i \mathfrak{J}_i(H_k)/\mathfrak{J}(H_k) \geq \log k - O_b(1)$. Indeed, in the notation of that proof, all we need is to take $T = b/\log k$, $\tau = 1/\log k$, and $c = 1/\log k + \alpha/\log^2 k$ with a fixed sufficiently negative α . Rescale by $R_k(1 - 1/\log^2 k)$. The resulting support satisfies $\sum_i t_i \leq R_k(1 - 1/\log^2 k) < R_k$ and $t_i \leq bR_k(1 - 1/\log^2 k)/\log k < \delta(\theta/(4 \log k))$ for every i . We then adapt [Pol14b, proof of Theorem 22, (72)–(84)] to approximate the rescaled density in L^2 by a finite sum of smooth tensor products. Each product can be supported in a rectangle $\prod_i [a_i, b_i]$ with $\sum_i b_i < R_k$ and $b_i < \delta(\theta/(4 \log k))$ for every i . Since the ratio is continuous under L^2 approximation on this bounded region, we obtain a density G_k with $\mathfrak{J}(G_k) = 1$ and

$$\sum_{i=1}^k \mathfrak{J}_i(G_k) \geq \frac{\theta}{2} \log k - C_2 \quad (4.1)$$

for some $C_2 > 0$ depending only on θ and δ . Define

$$F(t_1, \dots, t_k) := \int_{t_1}^{\infty} \dots \int_{t_k}^{\infty} G_k(u_1, \dots, u_k) du_k \dots du_1. \quad (4.2)$$

Then $F \in \mathcal{F}_k(\theta, \delta)$ and $\partial_{t_1 \dots t_k} F = (-1)^k G_k$.

Let k_* be large enough for (4.1) to hold, and choose

$$k = \max \left\{ k_*, \left\lceil \exp \left(\frac{2}{\theta \beta} (1 + \varepsilon/2)m + \frac{2C_2}{\theta} \right) \right\rceil \right\}. \quad (4.3)$$

For this fixed k and F , put

$$D := \beta \sum_{i=1}^k \mathfrak{J}_i(G_k) - (m-1)\mathfrak{J}(G_k) \geq 1 + \varepsilon m/2 > 0.$$

Fix $C_F > 0$ for the bound in Proposition 3.3, and choose

$$0 < \gamma \leq \min \left\{ \frac{1}{100k}, \frac{D}{4C_F k^2} \right\}.$$

Define S as in (3.1) with these choices. Combining Propositions 3.1, 3.2 and 3.3 gives

$$S \geq (D - C_F k^2 \gamma - o(1)) \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}. \quad (4.4)$$

Taking q large enough that $o(1) \leq D/4$, we obtain

$$S \geq \frac{D}{2} \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}. \quad (4.5)$$

Finally, we need to bound the weights w_a over $\Omega(a + h_i q) < 2/\gamma$ for all i . If $a \in (q, 2q]$ satisfies the bound on the number of prime factors $\Omega(a + qh_i) < z$ for all i , then

$$\frac{w_a}{\max_{(t_1, \dots, t_k) \in \Delta_k(\theta, \delta)} F(t_1, \dots, t_k)^2} \leq \left(\prod_{i=1}^k \sum_{d|a+qh_i} \mu^2(d) \right)^2 \leq 4^{zk}.$$

Since F is smooth and compactly supported, $\|F\|_\infty < \infty$. It follows that $M(z) \leq \|F\|_\infty^2 4^{kz}$ with $M(z)$ defined as in (3.2). Combining this with (3.3) and (4.5), we get

$$\#\{a \pmod{q} : \text{at least } m \text{ of the } a + qh_i \text{ are primes}\} \gg_{m, \varepsilon} \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}.$$

Finally, if $(a + qh_i)_{i=1}^k$ has at least m primes, it follows that $p_m(q, a) \leq q(2 + h_k)$. As explained in [May15], if we select $h_1, \dots, h_k$ to be the first k primes that are larger than k , then $(h_1, \dots, h_k)$ is an admissible k -tuple that satisfies $h_k \ll k \log k$. The theorem follows.

5. PROOF OF PROPOSITIONS 3.1, 3.2 & 3.3

By the definition of $\mathcal{F}_k(\theta, \delta)$, F is a linear combination of tensor products, i.e. there are constants $c_j \in \mathbb{R}$ and $f_{i,j} \in \mathcal{C}_c^\infty([0, \infty))$ for $1 \leq i \leq k$ and $1 \leq j \leq J$ such that

$$F(t_1, \dots, t_k) = \sum_{j=1}^J c_j \prod_{i=1}^k f_{i,j}(t_i). \quad (5.1)$$

Let us denote

$$\lambda_f(n) := \sum_{d|n} \mu(d) f\left(\frac{\log d}{\log q}\right).$$

Then our weight becomes

$$\begin{aligned} w_a &= 1_{a \equiv b_0 \pmod{W_q}} \left(\sum_{j=1}^J c_j \prod_{i=1}^k \lambda_{f_{i,j}}(a + qh_i) \right)^2 \\ &= 1_{a \equiv b_0 \pmod{W_q}} \sum_{j=1}^J \sum_{j'=1}^J c_j c_{j'} \prod_{i=1}^k \lambda_{f_{i,j}}(a + qh_i) \lambda_{f_{i,j'}}(a + qh_i). \end{aligned}$$

Proof of Proposition 3.1. By linearity, it suffices to estimate

$$T := \sum_{\substack{q < a \leq 2q \\ (a, q) = 1}} 1_{a \equiv b_0 \pmod{W_q}} \prod_{i=1}^k \lambda_{f_i}(a + qh_i) \lambda_{g_i}(a + qh_i) \quad (5.2)$$

for all $f_i, g_i \in \mathcal{C}_c^\infty([0, \infty))$ with $1 \leq i \leq k$, with both $\prod_{i=1}^k f_i(t_i)$ and $\prod_{i=1}^k g_i(t_i)$ being supported in $\Delta_k(\theta, \delta)$. After interchanging the order of summation, this becomes

$$T = \sum_{d_1, d'_1, \dots, d_k, d'_k \geq 1} \left(\prod_{i=1}^k \mu(d_i) \mu(d'_i) f_i \left(\frac{\log d_i}{\log q} \right) g_i \left(\frac{\log d'_i}{\log q} \right) \right) S(\mathbf{d}, \mathbf{d}'), \quad (5.3)$$

where $\mathbf{d} := (d_1, \dots, d_k)$, $\mathbf{d}' := (d'_1, \dots, d'_k)$ and

$$S(\mathbf{d}, \mathbf{d}') := \sum_{\substack{q < a \leq 2q \\ [d_i, d'_i] \mid a + qh_i, 1 \leq i \leq k \\ a \equiv b_0 \pmod{W_q}}} 1_{(a, q)=1}.$$

Since $(h_1, \dots, h_k)$ is fixed and $w = \log \log \log q$, we take q sufficiently large that $w > \max_{i \neq j} |h_i - h_j|$. Note that the sum $S(\mathbf{d}, \mathbf{d}')$ vanishes unless $q, W_q, [d_1, d'_1], \dots, [d_k, d'_k]$ are pairwise coprime. In this case, let $r_{W_q, \mathbf{d}, \mathbf{d}'} := W_q \prod_{i=1}^k [d_i, d'_i]$. It follows from the Chinese remainder theorem that there exists $c_{W_q, \mathbf{d}, \mathbf{d}'}$ such that

$$S(\mathbf{d}, \mathbf{d}') = \sum_{\substack{q < a \leq 2q \\ a \equiv c_{W_q, \mathbf{d}, \mathbf{d}'} \pmod{r_{W_q, \mathbf{d}, \mathbf{d}'}}}} 1_{(a, q)=1}$$

By Möbius inversion, we arrive at

$$S(\mathbf{d}, \mathbf{d}') = \sum_{e \mid q} \mu(e) \sum_{\substack{q < a \leq 2q \\ a \equiv c_{W_q, \mathbf{d}, \mathbf{d}'} \pmod{r_{W_q, \mathbf{d}, \mathbf{d}'}}}} 1_{e \mid a}$$

From our previous discussion, we know that $(q, r_{W_q, \mathbf{d}, \mathbf{d}'}) = 1$. Hence,

$$S(\mathbf{d}, \mathbf{d}') = \sum_{e \mid q} \mu(e) \left(\frac{q}{er_{W_q, \mathbf{d}, \mathbf{d}'}} + O(1) \right) = \frac{\varphi(q)}{r_{W_q, \mathbf{d}, \mathbf{d}'}} + O(\tau(q)). \quad (5.4)$$

Therefore, we define T_1 (the main term) and T_2 (the error term) as

$$T_1 := \frac{\varphi(q)}{W_q} \sum_{\substack{d_1, d'_1, \dots, d_k, d'_k \geq 1 \\ q, W_q, [d_i, d'_i] \text{ pairwise coprime}}} \prod_{i=1}^k \frac{\mu(d_i) \mu(d'_i)}{[d_i, d'_i]} f_i \left(\frac{\log d_i}{\log q} \right) g_i \left(\frac{\log d'_i}{\log q} \right),$$

and $T_2 := T - T_1$.

We rewrite T_1 as

$$T_1 = \frac{\varphi(q)}{W_q} \sum_{\substack{d_1, d'_1, \dots, d_k, d'_k \geq 1 \\ q_w, W, [d_i, d'_i] \text{ pairwise coprime}}} \prod_{i=1}^k \frac{\mu(d_i) \mu(d'_i)}{[d_i, d'_i]} f_i \left(\frac{\log d_i}{\log q} \right) g_i \left(\frac{\log d'_i}{\log q} \right),$$

where $q_w := \prod_{p \mid q, p > w} p$ and $W = \prod_{p \leq w} p$. Then, applying [Pol14b, Lemma 30], this becomes

$$\begin{aligned} T_1 &= (\mathfrak{I}(\mathbf{f}, \mathbf{g}) + o(1)) \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W)}{W} \times \frac{\varphi(q_w)}{q_w} \times \log q \right)^{-k} \\ &= (\mathfrak{I}(\mathbf{f}, \mathbf{g}) + o(1)) \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}, \end{aligned}$$

where $\mathbf{f} := (f_1, \dots, f_k)$, $\mathbf{g} := (g_1, \dots, g_k)$ and

$$\mathfrak{I}(\mathbf{f}, \mathbf{g}) := \prod_{i=1}^k \int_0^\infty f'_i(t_i) g'_i(t_i) dt_i.$$

Recall that the supports of $(t_1, \dots, t_k) \mapsto \prod_{i=1}^k f_i(t_i)$ and $(t_1, \dots, t_k) \mapsto \prod_{i=1}^k g_i(t_i)$ are both contained in $\Delta_k(\theta, \delta)$. We bound T_2 using (5.3) and (5.4):

$$\begin{aligned} T_2 &\ll_{f,g} \tau(q) \sum_{d_1, d'_1, \dots, d_k, d'_k \geq 1} \prod_{i=1}^k 1_{\sum_{i=1}^k \frac{\log d_i}{\log q} \leq \frac{\theta}{2} (1 - \frac{1}{2 \log k})} \cdot 1_{\sum_{i=1}^k \frac{\log d'_i}{\log q} \leq \frac{\theta}{2} (1 - \frac{1}{2 \log k})} \\ &= \tau(q) \cdot \left(\sum_{a \leq q^{\frac{\theta}{2} (1 - \frac{1}{2 \log k})}} \tau_k(a) \right)^2 \\ &\ll_{f,g,k} q^{\theta(1 - \frac{1}{3 \log k})} = o \left(\frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k} \right). \end{aligned}$$

The proposition follows. $\square$

Proof of Proposition 3.2. Since ρ is a minorant for the indicator of primes, we know that $\rho(a + qh_{i_0}, q) \leq 0$ if $(a, q) > 1$. Therefore,

$$\sum_{\substack{q < a \leq 2q \\ (a, q) = 1}} \rho(a + qh_{i_0}, q) w_a \geq \sum_{q < a \leq 2q} \rho(a + qh_{i_0}, q) w_a \quad (5.5)$$

For our analysis, we suppose $i_0 = k$ (the cases $i_0 < k$ are treated in the same way). If $\rho(a + qh_k, q) \neq 0$, then any divisor $d|a + qh_k$ must either satisfy $d = 1$ or $d \geq q^\xi$ by Definition 2.1 (ii). For all j, j' , since both $f_{k,j}$ and $f_{k,j'}$ are supported in $[0, \xi]$, we have $\lambda_{f_{k,j}}(a + qh_k) \lambda_{f_{k,j'}}(a + qh_k) = f_{k,j}(0) f_{k,j'}(0)$ whenever $\rho(a + qh_k, q) \neq 0$. Thus,

$$\sum_{q < a \leq 2q} \rho(a + qh_{i_0}, q) w_a = \sum_{j=1}^J \sum_{j'=1}^J c_j c_{j'} f_{k,j}(0) f_{k,j'}(0) \tilde{T}(j, j'), \quad (5.6)$$

with

$$\tilde{T}(j, j') := \sum_{\substack{q < a \leq 2q \\ a \equiv b_0 \pmod{W_q}}} \rho(a + qh_k, q) \prod_{i=1}^{k-1} \lambda_{f_{i,j}}(a + qh_i) \lambda_{f_{i,j'}}(a + qh_i).$$

For general $f_i, g_i \in \mathcal{C}_c^\infty([0, \infty))$ with $1 \leq i \leq k-1$, define

$$\tilde{T} := \sum_{\substack{q < a \leq 2q \\ a \equiv b_0 \pmod{W_q}}} \rho(a + qh_k, q) \prod_{i=1}^{k-1} \lambda_{f_i}(a + qh_i) \lambda_{g_i}(a + qh_i).$$

If we prove the asymptotic formula

$$\tilde{T} = \frac{\mathfrak{I}_k(\mathbf{f}, \mathbf{g}) + o(1)}{\varphi(W_q)} \left(\sum_{q < a \leq 2q} \rho(a + qh_k, q) \right) \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-(k-1)} \quad (5.7)$$

with

$$\mathfrak{I}_k(\mathbf{f}, \mathbf{g}) := \prod_{i=1}^{k-1} \int_0^\infty f'_i(t_i) g'_i(t_i) dt_i,$$

then, after recombining the finitely many tensor terms, the proposition follows from (5.5), (5.6), Property (iv) with $\ell = h_k + 1$ of Definition 2.1, and the identity

$$\sum_{j=1}^J \sum_{j'=1}^J c_j c_{j'} f_{k,j}(0) f_{k,j'}(0) \mathfrak{J}_k(\mathbf{f}_j, \mathbf{f}_{j'}) = \mathfrak{J}_k(\partial_{t_1 \dots t_k} F) \geq 0.$$

After interchanging the order of summation, we have

$$\tilde{T} = \sum_{d_1, d'_1, \dots, d_{k-1}, d'_{k-1} \geq 1} \left(\prod_{i=1}^{k-1} \mu(d_i) \mu(d'_i) f_i \left(\frac{\log d_i}{\log q} \right) g_i \left(\frac{\log d'_i}{\log q} \right) \right) \tilde{S}(\mathbf{d}, \mathbf{d}'),$$

where

$$\tilde{S}(\mathbf{d}, \mathbf{d}') := \sum_{\substack{q < a \leq 2q \\ [d_i, d'_i] \mid a + qh_k, 1 \leq i \leq k-1 \\ a \equiv b_0 \pmod{W_q}}} \rho(a + qh_k, q).$$

Arguing as in the proof of Proposition 3.1, there exists $c_{W_q, \mathbf{d}, \mathbf{d}'}$ such that

$$\tilde{S}(\mathbf{d}, \mathbf{d}') = \frac{1}{\varphi(\tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'})} \sum_{q < a \leq 2q} \rho(a + qh_k, q) + \Delta(q, h_k + 1, c_{W_q, \mathbf{d}, \mathbf{d}'} + qh_k, \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'}) \quad (5.8)$$

by the Chinese Remainder Theorem, with $\tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'} := W_q \prod_{i=1}^{k-1} [d_i, d'_i]$ and

$$\Delta(q, \ell, c, r) := \sum_{\ell q < a \leq (\ell+1)q} \rho(a, q) \left(1_{a \equiv c \pmod{r}} - \frac{1_{(a, r)=1}}{\varphi(r)} \right),$$

if q, W_q and all $[d_i, d'_i]$ are pairwise coprime. Note that $a + qh_k$ and $\tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'}$ are coprime whenever $\rho(a + qh_k, q) \neq 0$ since all prime factors of $\tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'}$ are $\leq q^\xi$, which is why we did not explicitly restrict the sum in (5.8) to a 's with $a + qh_k$ and $\tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'}$ being coprime. We define $\tilde{T}_1$ as

$$\begin{aligned} \tilde{T}_1 &:= \frac{1}{\varphi(W_q)} \left(\sum_{q < a \leq 2q} \rho(a + qh_k, q) \right) \\ &\quad \times \sum_{\substack{d_1, d'_1, \dots, d_{k-1}, d'_{k-1} \geq 1 \\ q, W_q, [d_i, d'_i] \text{ pairwise coprime}}} \prod_{i=1}^{k-1} \frac{\mu(d_i) \mu(d'_i)}{\varphi([d_i, d'_i])} f_i \left(\frac{\log d_i}{\log q} \right) g_i \left(\frac{\log d'_i}{\log q} \right), \end{aligned}$$

and $\tilde{T}_2 := \tilde{T} - \tilde{T}_1$. Note that

$$\begin{aligned} \tilde{T}_1 &= \frac{1}{\varphi(W_q)} \left(\sum_{q < a \leq 2q} \rho(a + qh_k, q) \right) \\ &\quad \times \sum_{\substack{d_1, d'_1, \dots, d_{k-1}, d'_{k-1} \geq 1 \\ q, W_q, [d_i, d'_i] \text{ pairwise coprime}}} \prod_{i=1}^{k-1} \frac{\mu(d_i) \mu(d'_i)}{\varphi([d_i, d'_i])} f_i \left(\frac{\log d_i}{\log q} \right) g_i \left(\frac{\log d'_i}{\log q} \right). \end{aligned}$$

where we recall from the proof of the previous proposition $q_w = \prod_{p|q, p>w} p$ and $W = \prod_{p \leq w} p$. Put $B_q := \frac{\varphi(W_q)}{W_q} \frac{\varphi(q)}{q} \log q$. We will prove that $\tilde{T}_2 = o\left(\frac{q}{\varphi(W_q) \log q} B_q^{-(k-1)}\right)$. Applying [Pol14b, Lemma 30], we have

$$\begin{aligned} \tilde{T}_1 &= \frac{\mathfrak{I}_k(\mathbf{f}, \mathbf{g}) + o(1)}{\varphi(W_q)} \left(\sum_{q < a \leq 2q} \rho(a + qh_k, q) \right) \left(\frac{\varphi(W)}{W} \times \frac{\varphi(q_w)}{q_w} \times \log q \right)^{-(k-1)} \\ &= \frac{\mathfrak{I}_k(\mathbf{f}, \mathbf{g}) + o(1)}{\varphi(W_q)} \left(\sum_{q < a \leq 2q} \rho(a + qh_k, q) \right) \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-(k-1)} \end{aligned}$$

To bound $\tilde{T}_2$, recall the product $P(z) := \prod_{p \leq z} p$. The product $\prod_{i=1}^{k-1} f_i\left(\frac{\log d_i}{\log q}\right) g_i\left(\frac{\log d'_i}{\log q}\right)$ vanishes unless $(\mathbf{d}, \mathbf{d}')$ belongs to the set

$$\mathcal{D} = \left\{ (\mathbf{d}, \mathbf{d}') \in \mathbb{N}^{k-1} \times \mathbb{N}^{k-1} : \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'} \leq q^{\theta - \varepsilon_k} \text{ and } \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'} | P(q^{\delta(\varepsilon_k)}) \right\}$$

for $\varepsilon_k := \frac{\theta}{4 \log k}$. We have the upper bound

$$\tilde{T}_2 \ll \sum_{(\mathbf{d}, \mathbf{d}') \in \mathcal{D}} \left| \Delta\left(q, h_k + 1, c_{W_q, \mathbf{d}, \mathbf{d}'} + qh_k, \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'}\right) \right|.$$

Let $\mathcal{C}$ be the set of positive integers $c \leq P(q^{\delta(\varepsilon_k)})$ satisfying $c \equiv b_0 + qh_k \pmod{W_q}$ and for all $p \in (w, q^{\delta(\varepsilon_k)}]$, we have $c \equiv q(h_k - h_i) \pmod{p}$ for some $i \in \{1, \dots, k-1\}$. We have

$$\tilde{T}_2 \ll \sum_{(\mathbf{d}, \mathbf{d}') \in \mathcal{D}} \max_{\substack{c \in \mathcal{C} \\ (c, \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'} = 1}} \left| \Delta\left(q, h_k + 1, c, \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'}\right) \right|$$

Let $\omega(n)$ count the distinct prime factors of n . Note that for any squarefree $r \geq 1$ satisfying $W_q | r$ and $(r, q) = 1$, we have

$$\begin{aligned} &\# \left\{ (\mathbf{d}, \mathbf{d}') \in \mathbb{N}^{k-1} \times \mathbb{N}^{k-1} : r = \tilde{r}_{W_q, \mathbf{d}, \mathbf{d}'} \right\} \\ &= \sum_{a_1 \cdots a_{k-1} = r/W_q} \prod_{i=1}^{k-1} \# \left\{ (d_i, d'_i) \in \mathbb{N} \times \mathbb{N} : a_i = [d_i, d'_i] \right\} = (3k-3)^{\omega(r/W_q)}. \end{aligned}$$

Therefore,

$$\tilde{T}_2 \ll \sum_{\substack{r \leq q^{\theta - \varepsilon_k} \\ r | P(q^{\delta(\varepsilon_k)}) \\ (r, q) = 1}} (3k-3)^{\omega(r)} \max_{\substack{c \in \mathcal{C} \\ (c, r) = 1}} \left| \Delta\left(q, h_k + 1, c, r\right) \right|. \quad (5.9)$$

For any given r satisfying $r | P(q^{\delta(\varepsilon_k)})$, $r \leq q^{\theta - \varepsilon_k}$, and $(r, q) = 1$, and for any given $c_0 \in \mathcal{C}$ coprime to r , we have the lower bound

$$\# \{c \in \mathcal{C} : c \equiv c_0 \pmod{r}\} \geq \frac{|\mathcal{C}|}{k^{\omega(r)}}$$

by the Chinese Remainder Theorem. Thus,

$$\begin{aligned} \max_{\substack{c \in \mathcal{C} \\ (c,r)=1}} \left| \Delta(q, h_k + 1, c, r) \right| &\leq \frac{k^{\omega(r)}}{|\mathcal{C}|} \sum_{\substack{c \in \mathcal{C} \\ (c,r)=1}} \left| \Delta(q, h_k + 1, c, r) \right| \\ &\ll \frac{q^{1/2}}{\varphi(r)^{1/2}} \cdot \frac{k^{\omega(r)}}{|\mathcal{C}|} \sum_{\substack{c \in \mathcal{C} \\ (c,r)=1}} \left| \Delta(q, h_k + 1, c, r) \right|^{1/2}, \end{aligned}$$

where the last inequality follows from the trivial bound $|\Delta(q, h_k + 1, c, r)| \ll q/\varphi(r)$ (we have this bound because of the property that ρ is bounded above and below over its domain).

Combining this with (5.9) and using the Cauchy-Schwarz inequality, we have

$$\begin{aligned} \tilde{T}_2 &\ll \frac{q^{1/2}}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \sum_{\substack{r \leq q^{\theta - \varepsilon_k} \\ r | P(q^{\delta(\varepsilon_k)}) \\ (r,q)=1 \\ (c,r)=1}} \frac{(3k(k-1))^{\omega(r)}}{\varphi(r)^{1/2}} \cdot \left| \Delta(q, h_k + 1, c, r) \right|^{1/2} \\ &\ll \frac{q^{1/2}}{|\mathcal{C}|} \left(\sum_{r \leq q} \frac{(9k^4)^{\omega(r)}}{\varphi(r)} \right)^{1/2} \times \sum_{c \in \mathcal{C}} \left(\sum_{\substack{r \leq q^{\theta - \varepsilon_k} \\ r | P(q^{\delta(\varepsilon_k)}) \\ (r,q)=1 \\ (c,r)=1}} \left| \Delta(q, h_k + 1, c, r) \right| \right)^{1/2} \ll \frac{q}{(\log q)^{k^4/2}} \end{aligned}$$

by applying Property (iii) of Definition 2.1 with $A = 10k^4$. Since $\frac{q}{(\log q)^{k^4/2}} = o\left(\frac{q}{\varphi(W_q) \log q} B_q^{-(k-1)}\right)$, this proves the required error estimate and hence (5.7). The proposition follows. $\square$

Proof of Proposition 3.3. Similarly, it suffices to estimate

$$\sum_{\substack{q < a \leq 2q \\ (a,q)=1}} \left(\sum_{\substack{p | a + qh_k \\ p \leq q^\gamma}} 1 \right) 1_{a \equiv b_0 \pmod{W_q}} \prod_{i=1}^k \lambda_{f_i}(a + qh_i) \lambda_{g_i}(a + qh_i). \quad (5.10)$$

If $p \mid q$, then $(a, q) = 1$ implies $p \nmid a + qh_k$, so these primes contribute nothing to (5.10). If $p \nmid W_q$, the choice of b_0 likewise excludes divisibility; if $p \nmid qW_q$, the condition $p \mid a + qh_k$ selects one residue class modulo p . The unmarked local factors therefore match those in Proposition 3.1, including its $\varphi(q)/q$ normalization. Adapting the proof of [Pol14b, Proposition 14 (Almost primality), pp. 23–27], one can show that

$$\begin{aligned} &\sum_{\substack{q < a \leq 2q \\ (a,q)=1}} 1_{p | a + qh_k} 1_{a \equiv b_0 \pmod{W_q}} \prod_{i=1}^k \lambda_{f_i}(a + qh_i) \lambda_{g_i}(a + qh_i) \\ &\ll \frac{\log p}{p \log q} \times \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k}. \end{aligned}$$

Then, by summing over primes p , the expression (5.10) is

$$\ll \gamma \times \frac{\varphi(q)}{W_q} \left(\frac{\varphi(W_q)}{W_q} \times \frac{\varphi(q)}{q} \times \log q \right)^{-k},$$

and the proposition follows. $\square$

6. PROOF OF THEOREM 2.2

Let $h := C \exp(3.8075m)$, where $C > 0$ is the absolute constant in Corollary 2.1. Then using Lemma 1.1 followed by the triangle inequality, there exists the least integer $1 \leq q \leq \mu(B(x_0; \varepsilon/4h))^{-1}$ such that

$$d(T^q x_0, x_0) < \frac{\varepsilon}{2h}.$$

Since x_0 is nonperiodic, the least q satisfying $d(T^q x_0, x_0) < \varepsilon/(2h)$ tends to infinity as $\varepsilon \rightarrow 0$, so for ε sufficiently small in terms of m and x_0 , Corollary 2.1 applies to this q . Let $\mathcal{A} := \{1 \leq a \leq q : p_m(q, a) \leq qh, (a, q) = 1\}$ and $\mathcal{X}_{\mathcal{A}} := \bigcup_{a \in \mathcal{A}} B(T^a x_0; \varepsilon/4h)$. Note that $\mathcal{X}_{\mathcal{A}}$ is in fact a disjoint union by the minimality of q . Then, for $a \in \mathcal{A}$, $x \in B(T^a x_0; \varepsilon/4h)$, and $1 \leq i \leq m$, we have

$$\begin{aligned} d(T^{p_i(q,a)} x_0, x) &\leq d(T^{p_i(q,a)} x_0, T^a x_0) + d(T^a x_0, x) \\ &< d(T^{p_i(q,a)} x_0, T^a x_0) + \frac{\varepsilon}{4h}. \end{aligned} \quad (6.1)$$

Since T is an isometry, we have

$$\begin{aligned} d(T^{p_i(q,a)} x_0, T^a x_0) &= d(T^{p_i(q,a)-a} x_0, x_0) \\ &\leq \sum_{1 \leq j \leq \frac{p_i(q,a)-a}{q}} d(T^{jq} x_0, T^{(j-1)q} x_0) \\ &= \frac{p_i(q,a) - a}{q} \times d(T^q x_0, x_0) \\ &< \frac{\varepsilon}{2}. \end{aligned}$$

Then, it follows from (6.1) that for all $x \in \mathcal{X}_{\mathcal{A}}$, there exists $a \in \mathcal{A}$ such that $d(T^{p_i(q,a)} x_0, x) < \varepsilon$ for all $i \in \{1, \dots, m\}$, and thus $p_m^T(x_0, x; \varepsilon) \leq qh$.

On the other hand, since $T^{-a}(B(T^a x_0; \varepsilon/4h)) = B(x_0; \varepsilon/4h)$ and T is measure-preserving, we have $\mu(B(T^a x_0; \varepsilon/4h)) = \mu(B(x_0; \varepsilon/4h))$, which implies that

$$\begin{aligned} \mu(\mathcal{X}_{\mathcal{A}}) &= \#\mathcal{A} \times \mu(B(x_0; \varepsilon/4h)) \\ &\gg_m \varphi(q)(1 + \log q)^{-C' \exp(3.8075m)} \times \mu(B(x_0; \varepsilon/4h)) \end{aligned}$$

by Corollary 2.1. Set $J = \lceil \log_2(4h) \rceil$; iterating pseudo-doubling gives $\mu(B(x_0; \varepsilon/4h)) \geq \lambda^{-J} \mu(B(x_0; \varepsilon))$ and $q \leq \lambda^J \mu(B(x_0; \varepsilon))^{-1} \ll_{\lambda} m^{\log_2 \lambda} \exp(3.8075m \log_2 \lambda) \mu(B(x_0; \varepsilon))^{-1}$. This proves the theorem.

To obtain Theorem 1.2 for any $x_0 \in \text{supp } \mu$, let $Q(m)$ be the modulus threshold in Corollary 2.1 and increase h to $h_m := \max\{h, \max_{1 \leq q < Q(m)} \min_{(a,q)=1} p_m(q, a)/q\}$, with the usual convention for $q = 1$. Choose the least return at radius $\varepsilon/(2h_m)$ as above. For $q \geq Q(m)$, Corollary 2.1 supplies a suitable residue class; for $q < Q(m)$, its existence follows from the definition of h_m . The same distance and pseudo-doubling estimates give Theorem 1.2 with an implied constant depending only on m and λ .

7. FIRST RETURN TIME

Given any $\varepsilon > 0$, $x_0 \in X$ and $x \in B(x_0; \varepsilon)$, let $\tau_{x_0; \varepsilon}(x) := \min\{n \geq 1 : T^n x \in B(x_0; \varepsilon)\}$ denote the *first return time*. For instance, the least integer $q \geq 1$ for which $d(T^q x_0, x_0) < \varepsilon/2h$ in the proof of Theorem 2.2 is the first return time $\tau_{x_0; \varepsilon/2h}(x_0)$. Suppose the measure-preserving map T is ergodic and $\mu(B(x_0; \varepsilon/2h)) > 0$. Then Kac's lemma [Kac47] states

that

$$\int_{B(x_0, \varepsilon/2h)} \tau_{x_0; \varepsilon/2h}(x) d\mu(x) = 1,$$

i.e., under the conditional probability measure $\mu(\cdot \mid B(x_0; \varepsilon/2h))$, the expected return time is $\mu(B(x_0, \varepsilon/2h))^{-1}$. For the following conditional estimate, suppose instead that x_0 is nonperiodic and its first return time satisfies

$$\mu(B(x_0; \varepsilon/2h))^{-1} (\log \mu(B(x_0; \varepsilon/2h))^{-1})^{-100} \ll \tau_{x_0; \varepsilon/2h}(x_0) \ll \mu(B(x_0; \varepsilon/2h))^{-1} \quad (7.1)$$

This is a separate hypothesis on the centre x_0 .

Suppose now (7.1) holds. Let $q' := \tau_{x_0; \varepsilon/2h}(x_0)$. Then the proof of Theorem 2.2 gives

$$\begin{aligned} \mu(\{x \in X : p_m^T(x_0, x; \varepsilon) \leq C \mu(B(x_0; \varepsilon))^{-1} m^{\log_2 \lambda} \exp(3.8075(1 + \log_2 \lambda)m)\}) \\ \gg_{m, \lambda} \varphi(q')(1 + \log q')^{-C' \exp(3.8075m)} \mu(B(x_0; \varepsilon)) \\ \gg_{m, \lambda} (\log \mu(B(x_0; \varepsilon))^{-1})^{-C'' \exp(3.8075m)} \end{aligned}$$

for some absolute constant $C'' > 0$.

Unfortunately, none of the circle rotations are even weakly mixing (see [EW11, p. 51] for instance). In fact, one can verify that the assumption (7.1) fails for those circle rotations $T_\alpha : x \mapsto x + \alpha$ with α of type $\eta_\alpha > 1$,² where

$$\eta_\alpha := \sup\{\theta > 0 : \liminf_{n \rightarrow \infty} n^\theta \|n\alpha\| = 0\}.$$

More precisely, let $\tau_\varepsilon(\alpha) := \min\{n \geq 1 : \|n\alpha\| < \varepsilon\}$. Then Choe–Seo [CS01] proved that

$$\liminf_{\varepsilon \rightarrow 0^+} \frac{\log \tau_\varepsilon(\alpha)}{\log \varepsilon^{-1}} = \eta_\alpha^{-1}.$$

Nevertheless, we are able to establish tight lower and upper bounds for the first return time for most $\alpha \in (0, 1)$ of type $\eta_\alpha \leq 1$, thereby strengthening [KS03, Theorem 3].

Proposition 7.1. *Let $\alpha \in (0, 1)$, $\delta > 0$ and $\varepsilon > 0$ sufficiently small (in terms of α and δ). Then*

(1) *for almost all $\alpha \in (0, 1)$ with respect to the Lebesgue measure, we have*

$$\varepsilon^{-1} (\log \varepsilon^{-1})^{-(1+\delta)} \ll_\delta \tau_\varepsilon(\alpha) \ll \varepsilon^{-1};$$

(2) *the irrational $\alpha \in (0, 1)$ has bounded partial quotients³ if and only if*

$$\tau_\varepsilon(\alpha) \asymp_\alpha \varepsilon^{-1}.$$

In particular, the equation (7.1) holds, allowing us to derive the following (metric) Diophantine approximation result on early visits at prime times.

Corollary 7.1. *Let $m \geq 2$ and $\alpha \in (0, 1)$. Then for any sufficiently small $\varepsilon > 0$ (in terms of m, α), there exist absolute constants $C, C' > 0$ such that*

(1) *for almost all $\alpha \in (0, 1)$ with respect to the Lebesgue measure μ , we have*

$$\mu(\{\beta \in [0, 1] : p_m^\alpha(\beta; \varepsilon) \leq C \varepsilon^{-1} m \exp(7.615m)\}) \gg_m (\log \varepsilon^{-1})^{-C' \exp(3.8075m)};$$

(2) *for $\alpha \in (0, 1)$ with bounded partial quotients, we also have*

$$\mu(\{\beta \in [0, 1] : p_m^\alpha(\beta; \varepsilon) \leq C \varepsilon^{-1} m \exp(7.615m)\}) \gg_{m, \alpha} (\log \varepsilon^{-1})^{-C' \exp(3.8075m)}.$$

²By the Borel–Cantelli lemma, the set of irrationals α of type $\eta_\alpha > 1$ has Lebesgue measure 0.

³An irrational α with continued fraction expansion $[a_0; a_1, a_2, \dots]$ is said to have bounded partial quotients if a_n is uniformly bounded for $n \geq 1$. They are badly approximable by rationals, and vice versa, i.e., there exists a constant $c > 0$ such that $|\alpha - \frac{p}{q}| > \frac{c}{q^2}$ for all $\frac{p}{q} \in \mathbb{Q}$. For instance, this includes quadratic irrationals but excludes the base of the natural logarithm, $e = [2; 1, 2, 1, 1, 4, 1, 1, 6, 1, 1, 8, \dots]$.

Proof of Proposition 7.1. We begin with the proof of the first part. The upper bound follows from the Dirichlet approximation theorem. Since $\sum_{n=2}^{\infty} 1/(n(\log n)^{1+\delta})$ converges, the Borel–Cantelli lemma gives, for almost all irrational $\alpha \in [0, 1]$ and all sufficiently large integers n ,

$$\|n\alpha\| > \frac{1}{n(\log n)^{1+\delta}}.$$

For such an α , put $n = \tau_{\varepsilon}(\alpha)$. As $\varepsilon \rightarrow 0$, we have $n \rightarrow \infty$, while Dirichlet’s theorem gives $n \leq 2\varepsilon^{-1}$ for sufficiently small ε . Thus

$$\varepsilon > \|n\alpha\| > \frac{1}{n(\log n)^{1+\delta}} \geq \frac{1}{n(\log(2\varepsilon^{-1}))^{1+\delta}},$$

which gives the claimed lower bound.

If α has bounded partial quotients, then $\|n\alpha\| \gg_{\alpha} n^{-1}$ for every $n \geq 1$. Taking $n = \tau_{\varepsilon}(\alpha)$ and using $\|n\alpha\| < \varepsilon$ gives $\tau_{\varepsilon}(\alpha) \gg_{\alpha} \varepsilon^{-1}$. The upper bound follows from Dirichlet’s theorem.

For the converse, let $\alpha \in (0, 1)$ be an irrational number with unbounded partial quotients and denote $q_{n_k} := \min\{q_n : q_{n+1} > kq_n\}$ for $k \geq 1$. Note that such q_{n_k} always exists. Take $\varepsilon_k := q_{n_{k+1}}^{-1}$ for $k \geq 1$. Then by definition

$$\tau_{\varepsilon_k}(\alpha) \leq q_{n_k} < \frac{q_{n_{k+1}}}{k} = \frac{\varepsilon_k^{-1}}{k}.$$

Therefore, it is impossible that $\tau_{\varepsilon}(\alpha) \gg_{\alpha} \varepsilon^{-1}$ as $\varepsilon \rightarrow 0^+$, and the proof is completed. $\square$

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